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ROCKSTOR NAS INSTALLATION DOCUMENTATION

CLOIF2 PROJECT

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CONTENT

Table of Figures	2
Information Questions	5
Internal Hardware Components	6
Internal Ports and Connectors	7
StaginG	9
Hardware Installation	9
Installation of Native SSD/HDDs	10
Installation of RAID Controller and RAID HDDs	12
Installation of Power Supplies	13
Connection to monitor, keyboard and mouse	14
BIOS management	14
BIOS Configuration	15
RAID controller BIOS configuration	15
RAID Controller Configuration	17
OS Installation	19
Preparing the installation media	20
Bootting from the usb stick	21
Refreshing Rockstor and updating	25
Using the Web Dashboard	26
Rack Installation	27
Connection to RMM	29
Share configuration	31
Creating the software raid	32
Creating the Users and Groups	34
Using the shares	37
ISCSI protocol	38

TABLE OF FIGURES

Table 1: External backside connectors	5
Table 2: External frontside connectors	5
Table 3: Internal hardware components	6
Table 4: Server board ports and connectors	7
Table 5: Passive midplane ports and connectors	8
Figure 1: Server internals - showing correctly seated RAM sticks	9
Figure 2: Server internals showing the final installation of RAM sticks	9
Figure 3: 3.5" HDD installed in drive caddy	10
Figure 4: The first drives being inserted into the bay	10
Figure 5: The under side of the SSD	10
Figure 6: 3.5" caddy with 2.5" for the boot SSD assembled	10
Figure 7: Server internals showing the SATA cables connected to the motherboard, raid controller and backplane.....	11
Figure 8: 3.5" HDDs installed in drive caddys for the hardware RAID	12
Figure 9: The SATA cable connected to the backplane.....	12
Figure 10: RAID controller with PCIE riser installed in the server	13
Figure 11: The RAID controller with the riser being prepared for install	13
Figure 12: The back side of the server with the bootable USB stick inserted	13
Figure 13: Both power supplies connected to provide redundancy, both LEDs are lit up green.....	13
Figure 14: Monitor, keyboard and mouse connected to the server	14
Figure 15: The splash screen of the first boot.....	14
Figure 16: The bios showing RAM errors	14
Figure 17: The IP configuration page in the BIOS	15
Figure 18: The RAID controller selection screen.....	15
Figure 19: The RAID controller settings with Ethernet Configuration selected	15
Figure 20: The RAID controller network settings configured correctly	16
Figure 21: The "Create RAID Set" page on the hardware RAID controller webpage creating a RAID set	17
Figure 22: The RAID "Create Volume Set" webpage with the correct Volume set selected and RAID 1 being created	17
Figure 23: The RAID overview on the webpage	18
Figure 24: First option on the Rockstor download page	20
Figure 25: Checking the iso hash with sha256sum	20
Figure 26: Flashing the ISO with "dd" on Linux	20

Figure 27: The bootloader of the USB stick with "Install" selected	21
Figure 28: The installer loading	21
Figure 29: Rockstors welcome screen.....	21
Figure 30: System locale selection.....	22
Figure 31: Keyboard layout selection	22
Figure 32: Rockstors License agreement	23
Figure 33: The IP address configured on the network page.....	23
Figure 34: Hostname set to the correct value	24
Figure 35: Updating Rockstor directly after installing via SSH	25
Figure 36: Firefox complaining about Rockstors Self-Signed certificate.....	26
Figure 37: Rockstors initial admin account creation on the webpage	26
Figure 38: First look of the Rockstors dashboard	26
Figure 39: Server installed with rails in the rack	27
Figure 40: Server connected with power and ethernet	27
Figure 41: Better cable management with more servers being installed	28
Figure 42: All servers installed	28
Figure 43: The server powered on	29
Figure 44: Login page of the servers RMM and the FRU page	29
Figure 45: System information on the RMM page	30
Figure 46: The RMM logs that were full and that we deleted	30
Figure 47: Creating the first pool.....	32
Figure 48: Creating a pool by selecting the 2 physically connected hdds	32
Figure 49: Software storage pool created	33
Figure 50: Details of the software raid	33
Figure 51: Creating a group for all the users called family	34
Figure 52: All the groups on the rockstor NAS showing the default "users" group and our newly created "family" group	34
Figure 53: Creating the first user of three.....	34
Figure 54: Creating a 10GB share for the "yona" user	35
Figure 55: The two private shares for "donny" and "yona" created	35
Figure 56: Enabling the samba service with "MYGROUP" as the Workgroup	35
Figure 57: Exporting a private share over SMB.....	36
Figure 58: After enabling SMB we can use our credentials to connect to our private share via the IP address	37
Figure 59: Accessing a private share over Windows	37

Figure 60: All shares listed	37
Figure 61: Installing the necessary package for iSCSI	38
Figure 62: Deleting the temporary share we created to make space for iSCSI	38
Figure 63: Creating a folder where the iSCSI drive will be mounted	38
Figure 64: Opening the iSCSI cli and configuring it the first time	40
Figure 65: Getting the Initiator ID of the client: Opening the iSCSI initiator properties	40
Figure 66: Configuring iSCSI via the terminal and fixing the remaining issues	41
Figure 67: iSCSI Initiator on the client with the newly created VDisk connected	41
Figure 68: Hard drive showing up like a native drive would in partition manager	41
Figure 69: Format the drive.....	42
Figure 70: Mount it into your local filesystem.....	42

INFORMATION QUESTIONS

Do some research about the topic NAS. What is it? What is it used for?

NAS stands for Network Attached Storage, it's used for accessing a storage over the network instead of through direct connections.

What is the difference between DAS, NAS and a SAN?

DAS, Directly Attached Storage, is directly accessed on hardware. The host sees the hard drive as a local disk. They're not accessible over the network.

NAS, Network Attached Storage, are files and folders shared over the network, these are file-level access so you can't configure the format or filesystem of the folder remotely.

SAN, Storage Area Network, shares a disk over the network providing block-level access to it, appearing as a local disk on a host, allowing the configuration of the format and filesystem of the disk remotely.

Explain why in a professional environment, compute (CPU/RAM) and storage are usually separated.

It's separated to allow for independent scaling and management.

Describe the server form factor and casing dimensions

The server is 19 Zoll (48.26 cm) in width and 2 U (8,89 cm) in height.

Describe all the external server ports and connectors

Backside

Connector	Purpose / Description	Quantity
USB Ports	USB 2.0	4
Ethernet Ports	Mainboard LAN	2
	RMM LAN	1
	RAID Controller LAN	1
Serial port	RJ-45 (Console)	1
Power supply ports	IEC C14	2

Table 1: External backside connectors

Frontside

Connector	Purpose / Description	Quantity
USB Ports	USB 2.0	1
Display port	VGA	1

Table 2: External frontside connectors

INTERNAL HARDWARE COMPONENTS¹

Hardware Component	Purpose / Description	Quantity
CPUs with heatsinks	Main processors installed with finned heatsinks.	2
RAM modules	Installed memory modules, at least 1 in each channel	8
Intel Server Board	Main server board of the system.	1
Riser Card Assembly	PCIe expansion riser assembly.	1
RAID Controller	Hardware RAID controller installed in the riser	1
Passive SATA/SAS Midplane	Midplane between the server board, backplane, and control panel.	1
Hot-swap SATA/SAS Backplane	Backplane for the drive bays.	1
Bridge Board	Connects the main board and passive midplane path.	1
Hard Drive Bays	Hard drive bays for installing SATA or SAS Hard drives	5
Internal SSD	SSD needed for the Rockstor installation.	1
SATA HDDs	Native SATA HDDs used for the software RAID.	2
SAS HDDs	SAS disks used for the hardware RAID.	2
Power Supplies	Redundant power supply modules.	2
System Fan Assembly	System fans for cooling the CPUs and Hardware RAID controller.	3
Power Supply Fans	Fans inside the PSUs.	2
Processor Air Duct	Guides airflow through CPUs and memory.	1
Control Panel	Front control/indicator panel.	1

Table 3: Internal hardware components

¹Manual used for reference:
https://downloads.bl4ckb0x.de/download.intel.com/support/motherboards/server/s5520ur/sb/E51243-006_SR2600UR_SR2625UR_SG.pdf

INTERNAL PORTS AND CONNECTORS

Server board (SR2600UR) components

Connector / Port	Purpose / Description	Quantity
Intel Adaptive Slot	280-pin slot for the riser extension.	1
DIMM Slots	Memory slots, 6 assigned to CPU 1 and 6 to CPU 2.	12
SATA Connectors (mainboard)	On-board SATA ports for internal storage devices.	6
RMM3 Header	Connector for the Intel Remote Management Module 3.	1
Front Panel Connector	Connects the server front panel controls and indicators.	1
Fan Board Connector	Connects the board to the fan board assembly.	1
CPU Fan Headers	Fan headers for CPU 1 and CPU 2 cooling fans.	2
System Fan Headers	Headers for system fan connections.	2
Main Power Connector	24-pin main power connector for the server board.	1
CPU Power Connector	2x4 / 8-pin CPU power connector.	1
Power Supply SMBus Connector	Management/monitoring connection for the power supply.	1
USB Header	Internal USB header for additional USB connectivity.	1
SGPIO Header	Storage/backplane signalling connector.	1
I/O Module Mezzanine Connectors	Connectors for optional I/O module expansion.	2
Serial Port B Header	Internal serial header.	1
Bridge Board Connector	Connects the main board to the chassis bridge board / midplane path.	1

Table 4: Server board ports and connectors

Passive midplane components

Connector / Port	Purpose / Description	Quantity
Midplane Power Connector	Supplies power to the passive midplane.	1
SATA Connectors	Storage connectors on the passive midplane	8
Fan Power Connectors	Fan power connections	6
Bridge Board Connector	Connects the passive board to the bridge board	1
Backplane Connector	Connector for the hot-swap backplane	1

Table 5: Passive midplane ports and connectors

Installation of RAM Sticks



Figure 1: Server internals - showing correctly seated RAM sticks

Installed modules:

- 8x 2GB RAM sticks
 - Every stick is installed into each channel with each CPU having 1 channel with 2 sticks.
 - Due to some errors the RAM sticks had to be switched from slot 2 to slot 1 of each channel.

Final RAM installation



Figure 2: Server internals showing the final installation of RAM sticks

INSTALLATION OF NATIVE SSD/HDDS



Figure 6: 3.5" caddy with 2.5" for the boot SSD assembled



Figure 5: The under side of the SSD



Figure 3: 3.5" HDD installed in drive caddy



Figure 4: The first drives being inserted into the bay

Installed modules:

- 1x 256 GB SSD to install Rockstor onto
- 2x 160GB SATA HDDs to configure a software RAID 1

SATA Cable connections



Figure 7: Server internals showing the SATA cables connected to the motherboard, raid controller and backplane

INSTALLATION OF RAID CONTROLLER AND RAID HDDS

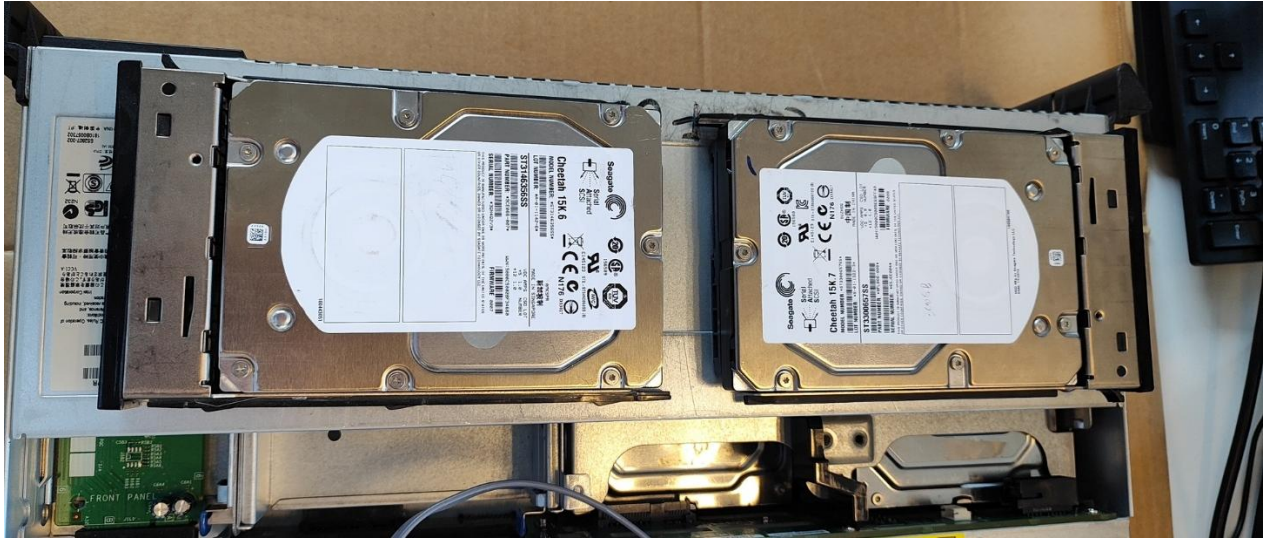


Figure 8: 3.5" HDDs installed in drive caddys for the hardware RAID

Installed modules:

2x 156 GB SAS HDDs

Hardware RAID Controller Installation



Figure 9: The SATA cable connected to the backplane



Figure 11: The RAID controller with the riser being prepared for install



Figure 10: RAID controller with PCIe riser installed in the server

Final SATA and SAS Cable connections:

Slot 1 – 3: SATA connected to mainboard

Slot 4 & 5 : SAS connected to RAID controller

INSTALLATION OF POWER SUPPLIES



Figure 12: The back side of the server with the bootable USB stick inserted

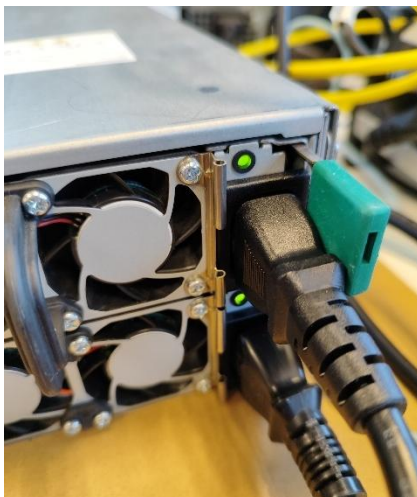


Figure 13: Both power supplies connected to provide redundancy, both LEDs are lit up green

CONNECTION TO MONITOR, KEYBOARD AND MOUSE



Figure 14: Monitor, keyboard and mouse connected to the server

BIOS MANAGEMENT

Hardware BIOS check & Errors



Figure 15: The splash screen of the first boot

RAM Error which made use switch from slot 2 to slot 1

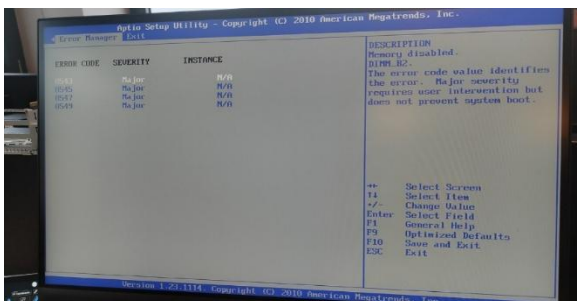


Figure 16: The bios showing RAM errors

After switching the RAM sticks to slot 1 of each channel this got fixed

BIOS CONFIGURATION

IP Configuration of the RMM.

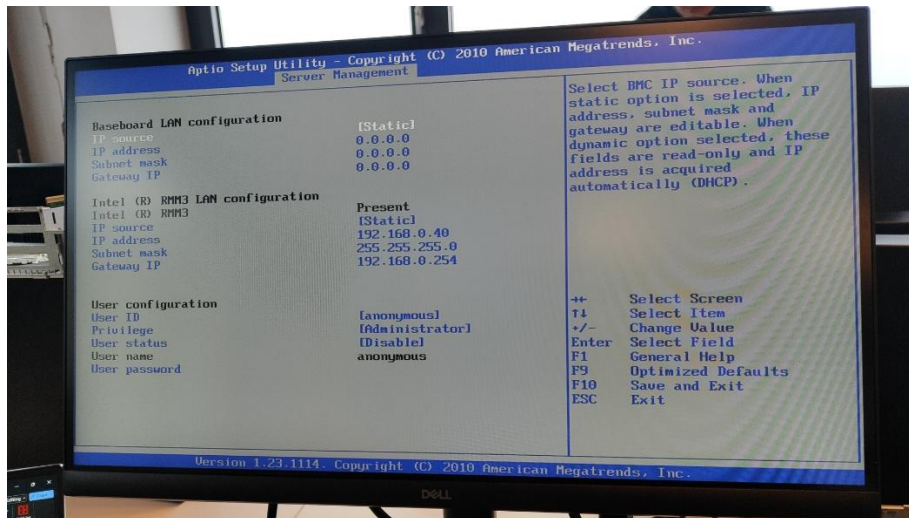


Figure 17: The IP configuration page in the BIOS

The rest of the BIOS can be left alone.

RAID CONTROLLER BIOS CONFIGURATION

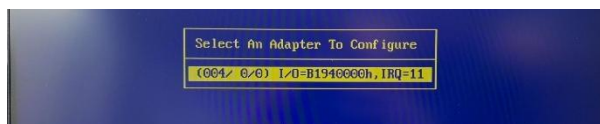


Figure 18: The RAID controller selection screen

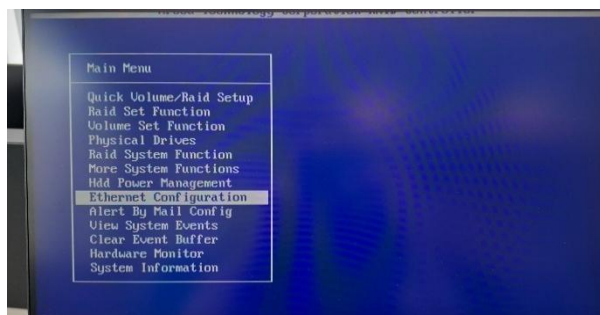


Figure 19: The RAID controller settings with Ethernet Configuration selected

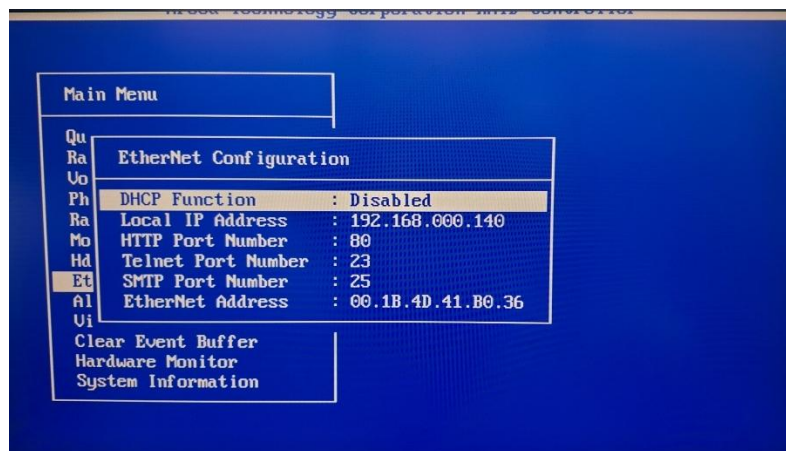


Figure 20: The RAID controller network settings configured correctly

RAID CONTROLLER CONFIGURATION

Connection to RAID controller web interface

Creation of a RAID Set using the 2 SAS HDDs connected to the controller

Areca Technology Corporation

[open all][close all]

Raid System Console
Quick Function
RAID Set Functions
Create RAID Set
Delete RAID Set
Expand RAID Set
Offline RAID Set
Rename RAID Set
Activate Incomplete RAID Set
Create Hot Spare
Delete Hot Spare
Rescue Raid Set
Volume Set Functions
Physical Drives
System Controls
Information

■ Select The Drives For RAID Set

Enclosure#1 : ARECA SAS RAID AdapterV1.0

	Slot#	Capacity	Model
☒	Slot#1	146.8GB	SEAGATE ST3146356SS
☒	Slot#4	146.8GB	SEAGATE ST3146356SS

Raid Set Name: Raid Set # 000
RaidSet Mode: Max 128 Volumes

☒ Confirm The Operation
Submit Reset

Figure 21: The "Create RAID Set" page on the hardware RAID controller webpage creating a RAID set

Volume Set creation

Areca Technology Corporation

[open all][close all]

Raid System Console
Quick Function
RAID Set Functions
Volume Set Functions
Create Volume Set
Create Raid30/50/60
Delete Volume Set
Modify Volume Set
Check Volume Set
Schedule Volume Check
Stop Volume Check
Physical Drives
System Controls
Information

■ Select The Raid Set To Create Volume On It

Select	Raid Set Name	Member Disks	Raid State	Capacity
☒	Raid Set # 000	2/2	Normal	280.0GB

Submit Reset

Areca Technology Corporation

[open all][close all]

Raid System Console
Quick Function
RAID Set Functions
Volume Set Functions
Create Volume Set
Create Raid30/50/60
Delete Volume Set
Modify Volume Set
Check Volume Set
Schedule Volume Check
Stop Volume Check
Physical Drives
System Controls
Information

■ Enter The Volume Attribute

Volume Name: areca_raid1
Member Disks: 2
Volume Raid Level: Raid 1
Max Capacity Allowed: 140 GB
Select Volume Capacity: 140 GB
Greater Two TB Volume Support: 64bit LBA
Volume Initialization Mode: No Initialization
Volume Stripe Size: 64 kBytes
Volume Cache Mode: Write Back
Volume Write Protection: Disabled
Tagged Command Queuing: Enabled
SCSI Channel:SCSI ID:SCSI Lun: 0 : 0 : 0
Volumes To Be Created: 1

☒ Confirm The Operation
Submit Reset

Figure 22: The RAID "Create Volume Set" webpage with the correct Volume set selected and RAID 1 being created

We only make a RAID 1 instead of 5 since we have only have 2 HDDs

Confirmation of RAID Set

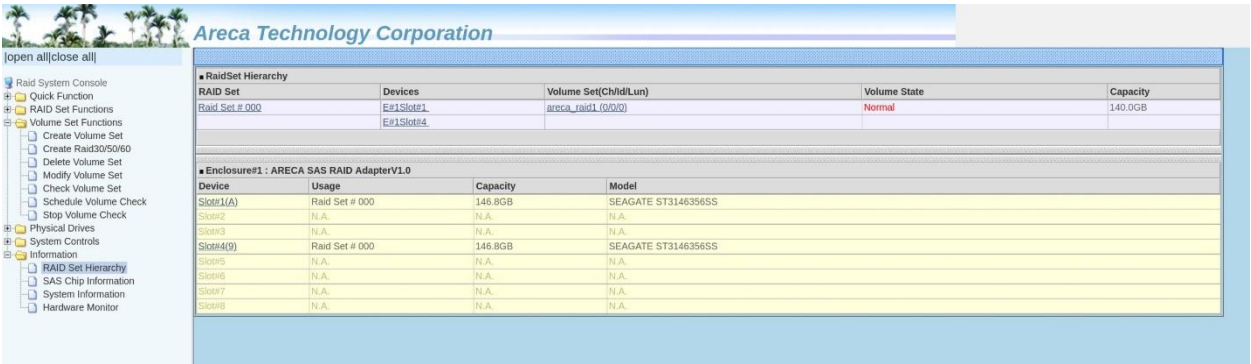


Figure 23: The RAID overview on the webpage

OS INSTALLATION

Description of NAS software

What is Rockstor?

Rockstor is a Linux and BTRFS powered storage operating system with advanced NAS and Cloud Storage features, supporting multiple architectures. It is based on openSUSE Linux and it offers a WebUI with a bunch of other features like docker.

Who created Rockstor?

Rockstor was founded by Suman Chakravartula as an Open Collective Non-Profit open source community endeavour. It is being maintained by Philip Paul Guyton and Flox. Dan W is in charge of maintaining Rock-ons.

Which is the latest release?

The latest release of Rockstor is 5.1.0-0, that is on the free “Testing Updates” tier.

How much does Rockstor cost?

Rockstor has two different tiers, a free tier and a paid tier. The paid tier offers stable tested updates (with high priority to fix bugs) and paid support is available whereas the free tier has the same feature set but is more on the bleeding edge and potentially less stable.

A stable subscription is available for 24€ a year on [opencollective.com](https://opencollective.com/rockstor).

What are the most important features?

- Because Rockstor is built entirely on the BTRFS file system, it supports advanced Copy-on-Write (CoW) capabilities and Bitrot protection to ensure high data integrity. System administrators can dynamically grow or shrink storage pools on the fly by adding or removing mixed-size drives without service interruption. It also allows users to switch between different RAID profiles online as their parity and performance needs change.
- The BTRFS foundation enables instantaneous, low-overhead file system snapshots that can be created manually or on an automated schedule. These snapshots can be cloned, rolled back in case of configuration errors, or exported directly to clients over the network. Rockstor also supports appliance-to-appliance replication to easily duplicate data across multiple NAS servers for disaster recovery.
- SMB/CIFS, NFS, Apple Filing Protocol (AFP), and SFTP are natively supported as share options out of the box. Additionally, it offers iSCSI support for block-level storage and integrates seamlessly with Active Directory and LDAP for enterprise user management.
- Hardware Flexibility allows the NAS software to be installed on basically any platform that supports Linux. Supported platforms, as of now, are x86, ARM64EFI and RaspberryPi4.

Why you should use Rockstor:

It uses BTRFS which is a new file system that carries a lot of benefits with it like instant snapshots and copy on write. Rockstor is built with that file system as the focus. It is also easy to install and manage as we found out by using it.

And why not?

Rockstor is a smaller project that might have less support than other NAS software alternatives like TrueNAS, OpenMediaVault or Unraid.

PREPARING THE INSTALLATION MEDIA

Connect your USB stick to your machine. Note down the name of the device and verify it's the correct device. This will erase all data so it is very important to pay attention.

Downloading the ISO image:

Go to the official Rockstor page and download the correct ISO for your platform.

<https://rockstor.com/dls/index.html>

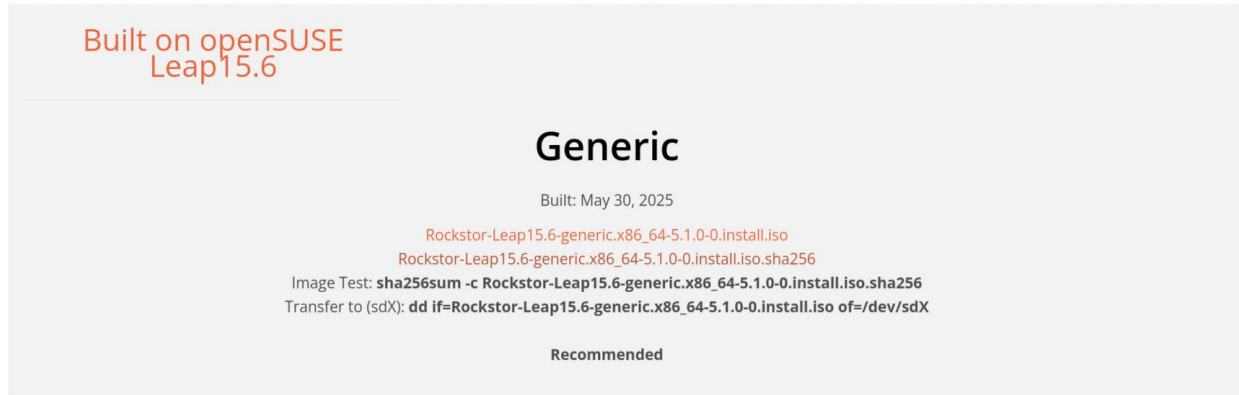


Figure 24: First option on the Rockstor download page

Usually it will be “Generic”. Compare the hash with the ISO to make sure it hasn’t been tampered with or is corrupt! This is especially important if you plan to use this in a production environment.

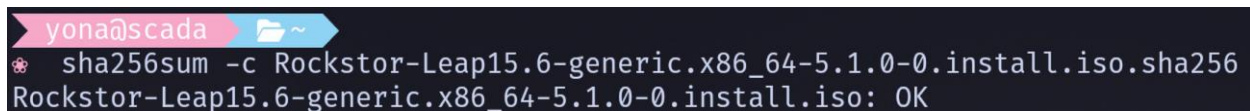


Figure 25: Checking the iso hash with sha256sum

Only continue if the ISO is verified.

Flashing the USB stick

Flash the USB stick with the ISO using your preferred method. In this case we used “dd” but you can also use “cat”, “cp”, or bigger programs like “Balena Etcher” and “Rufus”.



Figure 26: Flashing the ISO with "dd" on Linux

BOOTING FROM THE USB STICK

After selecting the correct boot option during the startup process, it should begin to boot from the USB stick. It will show a simple bootloader, choose the "Install" option and continue through the setup.

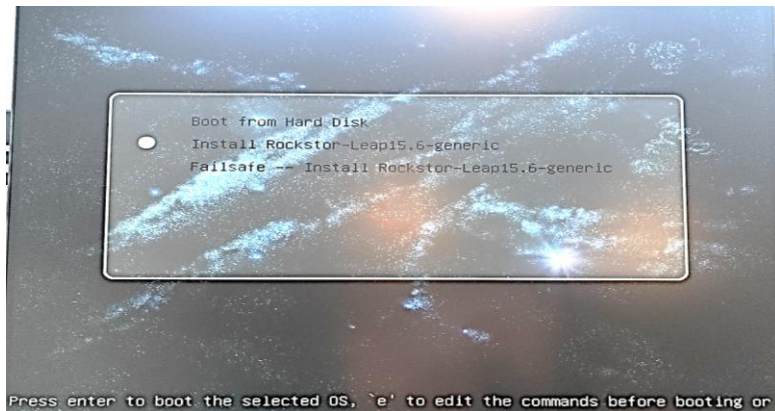


Figure 27: The bootloader of the USB stick with "Install" selected

It will now continue to load the installer to RAM.

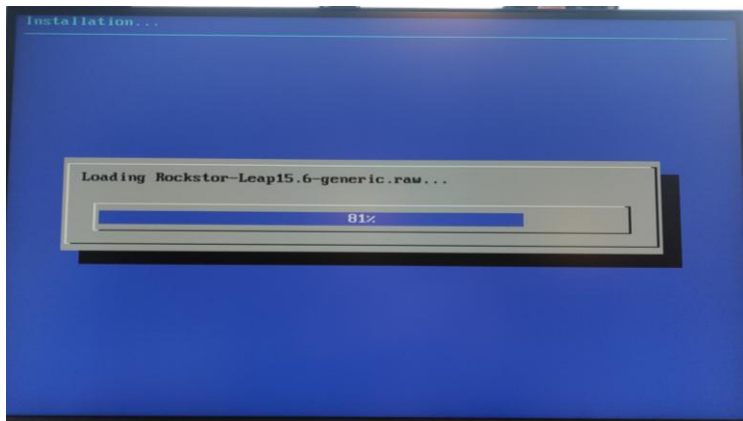


Figure 28: The installer loading



Figure 29: Rockstors welcome screen

Select the system locale, keyboard layout, accept the license, configure the network and hostname. After everything has been configured, install the OS to the SSD.

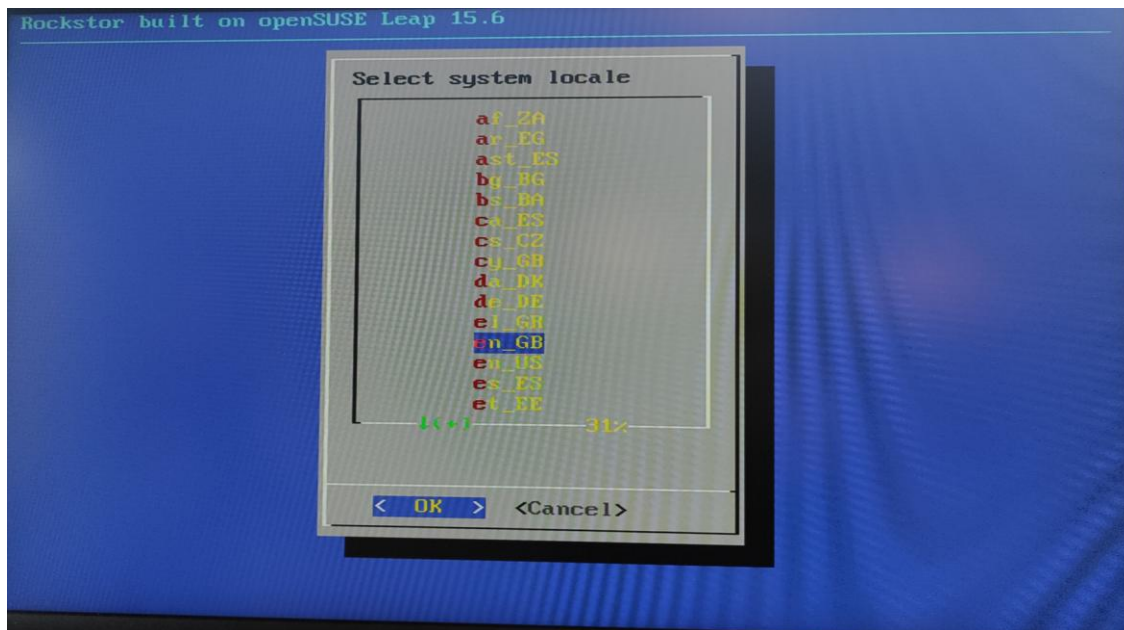


Figure 30: System locale selection



Figure 31: Keyboard layout selection

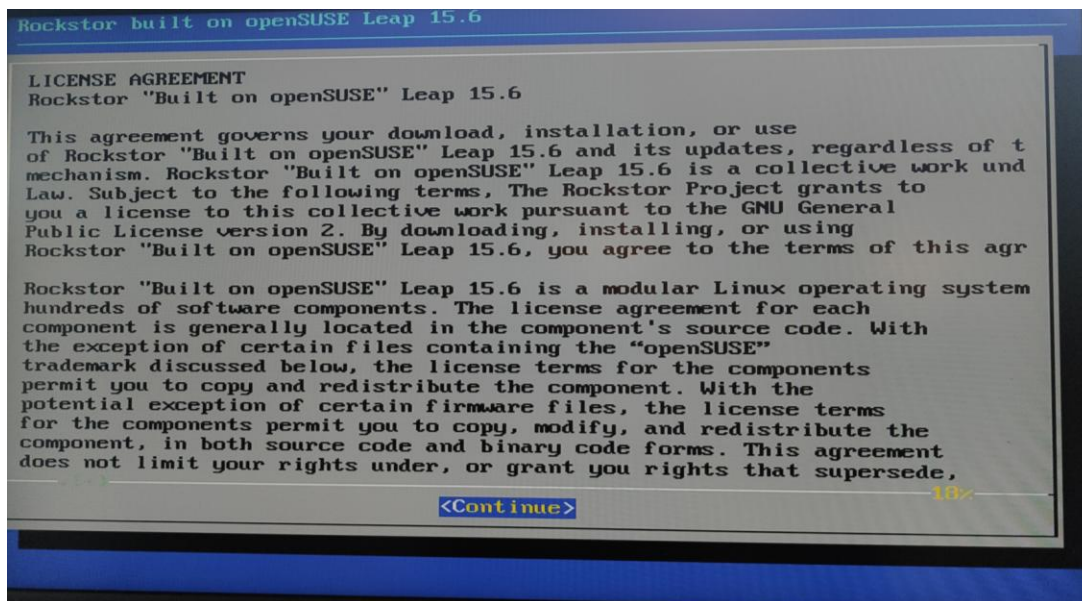


Figure 32: Rockstor's License agreement

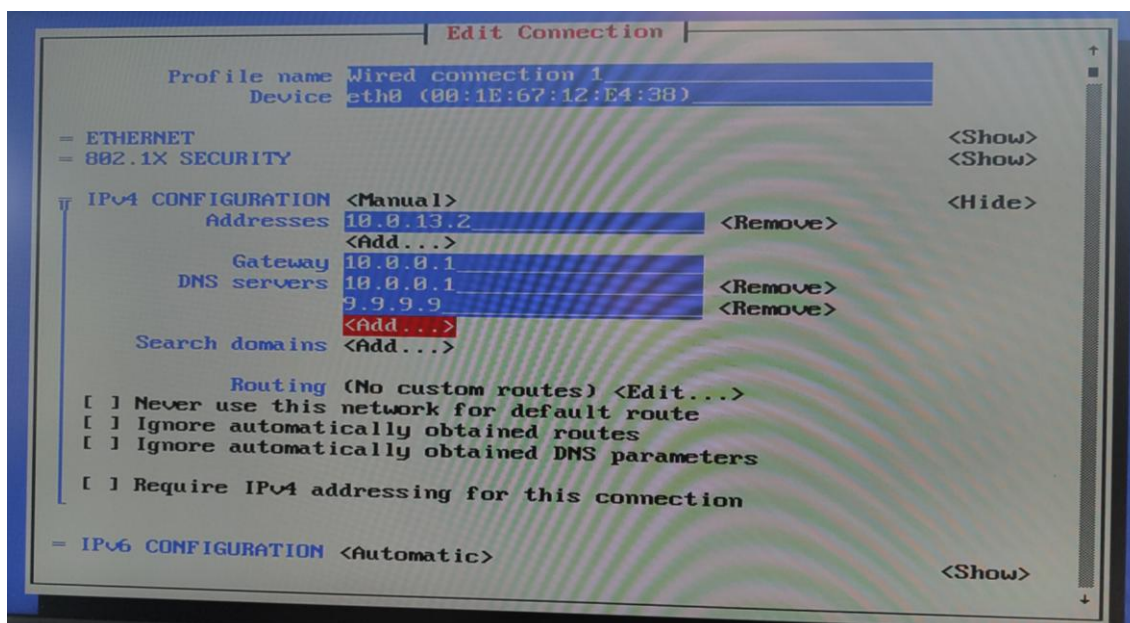


Figure 33: The IP address configured on the network page

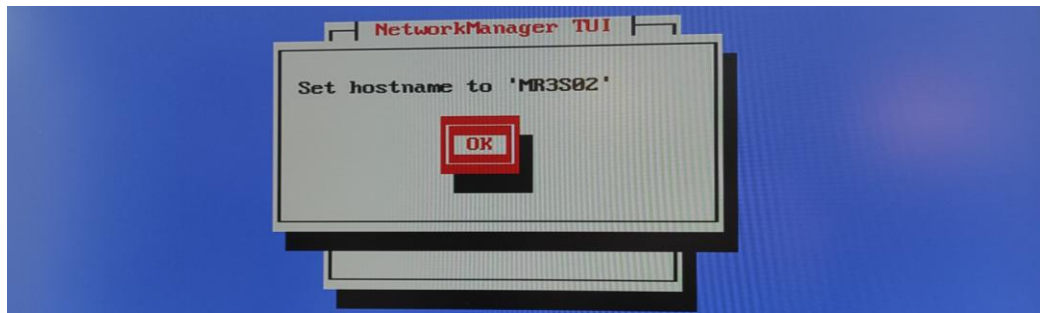


Figure 34: Hostname set to the correct value

REFRESHING ROCKSTOR AND UPDATING

After Rockstor is installed, we are able to connect to it over SSH. Update the repos and upgrade the packages.

```

* ssh root@10.0.13.2
The authenticity of host '10.0.13.2 (10.0.13.2)' can't be established.
ED25519 key fingerprint is: SHA256:BRNWhojpJlOdpH2GLj+VGNxSkS8Dag6J50R9z7owsUo
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.13.2' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
(root@10.0.13.2) Password:

Welcome to Rockstor built on openSUSE.
To update your system from here do:

    zypper refresh
    zypper up --no-recommends

Or use Rockstor's Web User Interface (Web-UI): recommended method.

Type 'myip' then Enter key to get my address for your Browser.
Enjoy ...
Last login: Wed Feb 25 10:40:39 2026
MR3S02:~ # myip
https://10.0.13.2
MR3S02:~ # zypper refresh
Retrieving repository 'Leap 15.6_Updates' metadata .....[done]
Building repository 'Leap 15.6_Updates' cache .....[done]
Retrieving repository 'Update repository of openSUSE Backports' metadata .....[done]
Building repository 'Update repository of openSUSE Backports' cache .....[done]
Retrieving repository 'Update repository with updates from SUSE Linux Enterprise 15' metadata .....[done]
Building repository 'Update repository with updates from SUSE Linux Enterprise 15' cache .....[done]
Repository 'Leap 15.6' is up to date.
Retrieving repository 'Virtualization Appliances Builder' metadata .....[done]
Repository 'home_rockstor' is up to date.
Repository 'home_rockstor_branches_Base_System' is up to date.
Retrieving repository 'repo-openh264' metadata .....[done]
Building repository 'repo-openh264' cache .....[done]
All repositories have been refreshed.
MR3S02:~ # zypper up --no-recommends
Loading repository data ...
Reading installed packages ...

The following 226 packages are going to be upgraded:
NetworkManager NetworkManager-branding-upstream NetworkManager-tui adaptec-firmware avahi bash bash-sh bind-utils binutils boost-license1_66_0
checkmedia chrony chrony-pool-openSUSE cifs-utils containerd coreutils crypto-policies cups-curl cyrus-sasl cyrus-sasl-gssapi
cyrus-sasl-plan device-mapper docker dracut dracut-kiwi-lib dracut-kiwi-oem-dump dracut-kiwi-oem-repart elfutils firewalld glib2-devel glib2-tools
glibc glibc-devel libcc-1 libcrypto-devel libcrypt2_0 libgio-2_0 libglib-2_0 libgmodule-2_0 libgnutls30 libgobject-2_0 libgomp1 libgpgme11
libgryptype6 libgcc_s1 libgcrpt2_0 libio-2_0 libl2_0 libl3_0 libl4_0 libl5_0 libl6_0 libl7_0 libl8_0 libl9_0 libl10_0 libl11_0 libl12_0 libl13_0 libl14_0 libl15_0 libl16_0 libl17_0 libl18_0 libl19_0 libl20_0 libl21_0 libl22_0 libl23_0 libl24_0 libl25_0 libl26_0 libl27_0 libl28_0 libl29_0 libl30_0 libl31_0 libl32_0 libl33_0 libl34_0 libl35_0 libl36_0 libl37_0 libl38_0 libl39_0 libl40_0 libl41_0 libl42_0 libl43_0 libl44_0 libl45_0 libl46_0 libl47_0 libl48_0 libl49_0 libl50_0 libl51_0 libl52_0 libl53_0 libl54_0 libl55_0 libl56_0 libl57_0 libl58_0 libl59_0 libl60_0 libl61_0 libl62_0 libl63_0 libl64_0 libl65_0 libl66_0 libl67_0 libl68_0 libl69_0 libl70_0 libl71_0 libl72_0 libl73_0 libl74_0 libl75_0 libl76_0 libl77_0 libl78_0 libl79_0 libl80_0 libl81_0 libl82_0 libl83_0 libl84_0 libl85_0 libl86_0 libl87_0 libl88_0 libl89_0 libl90_0 libl91_0 libl92_0 libl93_0 libl94_0 libl95_0 libl96_0 libl97_0 libl98_0 libl99_0 libl100_0 libl101_0 libl102_0 libl103_0 libl104_0 libl105_0 libl106_0 libl107_0 libl108_0 libl109_0 libl110_0 libl111_0 libl112_0 libl113_0 libl114_0 libl115_0 libl116_0 libl117_0 libl118_0 libl119_0 libl120_0 libl121_0 libl122_0 libl123_0 libl124_0 libl125_0 libl126_0 libl127_0 libl128_0 libl129_0 libl130_0 libl131_0 libl132_0 libl133_0 libl134_0 libl135_0 libl136_0 libl137_0 libl138_0 libl139_0 libl140_0 libl141_0 libl142_0 libl143_0 libl144_0 libl145_0 libl146_0 libl147_0 libl148_0 libl149_0 libl150_0 libl151_0 libl152_0 libl153_0 libl154_0 libl155_0 libl156_0 libl157_0 libl158_0 libl159_0 libl160_0 libl161_0 libl162_0 libl163_0 libl164_0 libl165_0 libl166_0 libl167_0 libl168_0 libl169_0 libl170_0 libl171_0 libl172_0 libl173_0 libl174_0 libl175_0 libl176_0 libl177_0 libl178_0 libl179_0 libl180_0 libl181_0 libl182_0 libl183_0 libl184_0 libl185_0 libl186_0 libl187_0 libl188_0 libl189_0 libl190_0 libl191_0 libl192_0 libl193_0 libl194_0 libl195_0 libl196_0 libl197_0 libl198_0 libl199_0 libl200_0 libl201_0 libl202_0 libl203_0 libl204_0 libl205_0 libl206_0 libl207_0 libl208_0 libl209_0 libl210_0 libl211_0 libl212_0 libl213_0 libl214_0 libl215_0 libl216_0 libl217_0 libl218_0 libl219_0 libl220_0 libl221_0 libl222_0 libl223_0 libl224_0 libl225_0 libl226_0 libl227_0 libl228_0 libl229_0 libl230_0 libl231_0 libl232_0 libl233_0 libl234_0 libl235_0 libl236_0 libl237_0 libl238_0 libl239_0 libl240_0 libl241_0 libl242_0 libl243_0 libl244_0 libl245_0 libl246_0 libl247_0 libl248_0 libl249_0 libl250_0 libl251_0 libl252_0 libl253_0 libl254_0 libl255_0 libl256_0 libl257_0 libl258_0 libl259_0 libl260_0 libl261_0 libl262_0 libl263_0 libl264_0 libl265_0 libl266_0 libl267_0 libl268_0 libl269_0 libl270_0 libl271_0 libl272_0 libl273_0 libl274_0 libl275_0 libl276_0 libl277_0 libl278_0 libl279_0 libl280_0 libl281_0 libl282_0 libl283_0 libl284_0 libl285_0 libl286_0 libl287_0 libl288_0 libl289_0 libl290_0 libl291_0 libl292_0 libl293_0 libl294_0 libl295_0 libl296_0 libl297_0 libl298_0 libl299_0 libl300_0 libl301_0 libl302_0 libl303_0 libl304_0 libl305_0 libl306_0 libl307_0 libl308_0 libl309_0 libl310_0 libl311_0 libl312_0 libl313_0 libl314_0 libl315_0 libl316_0 libl317_0 libl318_0 libl319_0 libl320_0 libl321_0 libl322_0 libl323_0 libl324_0 libl325_0 libl326_0 libl327_0 libl328_0 libl329_0 libl330_0 libl331_0 libl332_0 libl333_0 libl334_0 libl335_0 libl336_0 libl337_0 libl338_0 libl339_0 libl340_0 libl341_0 libl342_0 libl343_0 libl344_0 libl345_0 libl346_0 libl347_0 libl348_0 libl349_0 libl350_0 libl351_0 libl352_0 libl353_0 libl354_0 libl355_0 libl356_0 libl357_0 libl358_0 libl359_0 libl360_0 libl361_0 libl362_0 libl363_0 libl364_0 libl365_0 libl366_0 libl367_0 libl368_0 libl369_0 libl370_0 libl371_0 libl372_0 libl373_0 libl374_0 libl375_0 libl376_0 libl377_0 libl378_0 libl379_0 libl380_0 libl381_0 libl382_0 libl383_0 libl384_0 libl385_0 libl386_0 libl387_0 libl388_0 libl389_0 libl390_0 libl391_0 libl392_0 libl393_0 libl394_0 libl395_0 libl396_0 libl397_0 libl398_0 libl399_0 libl400_0 libl401_0 libl402_0 libl403_0 libl404_0 libl405_0 libl406_0 libl407_0 libl408_0 libl409_0 libl410_0 libl411_0 libl412_0 libl413_0 libl414_0 libl415_0 libl416_0 libl417_0 libl418_0 libl419_0 libl420_0 libl421_0 libl422_0 libl423_0 libl424_0 libl425_0 libl426_0 libl427_0 libl428_0 libl429_0 libl430_0 libl431_0 libl432_0 libl433_0 libl434_0 libl435_0 libl436_0 libl437_0 libl438_0 libl439_0 libl440_0 libl441_0 libl442_0 libl443_0 libl444_0 libl445_0 libl446_0 libl447_0 libl448_0 libl449_0 libl450_0 libl451_0 libl452_0 libl453_0 libl454_0 libl455_0 libl456_0 libl457_0 libl458_0 libl459_0 libl460_0 libl461_0 libl462_0 libl463_0 libl464_0 libl465_0 libl466_0 libl467_0 libl468_0 libl469_0 libl470_0 libl471_0 libl472_0 libl473_0 libl474_0 libl475_0 libl476_0 libl477_0 libl478_0 libl479_0 libl480_0 libl481_0 libl482_0 libl483_0 libl484_
```

Figure 35: Updating Rockstor directly after installing via SSH

USING THE WEB DASHBOARD

We can also connect to the WebUI by navigating to our IP address.

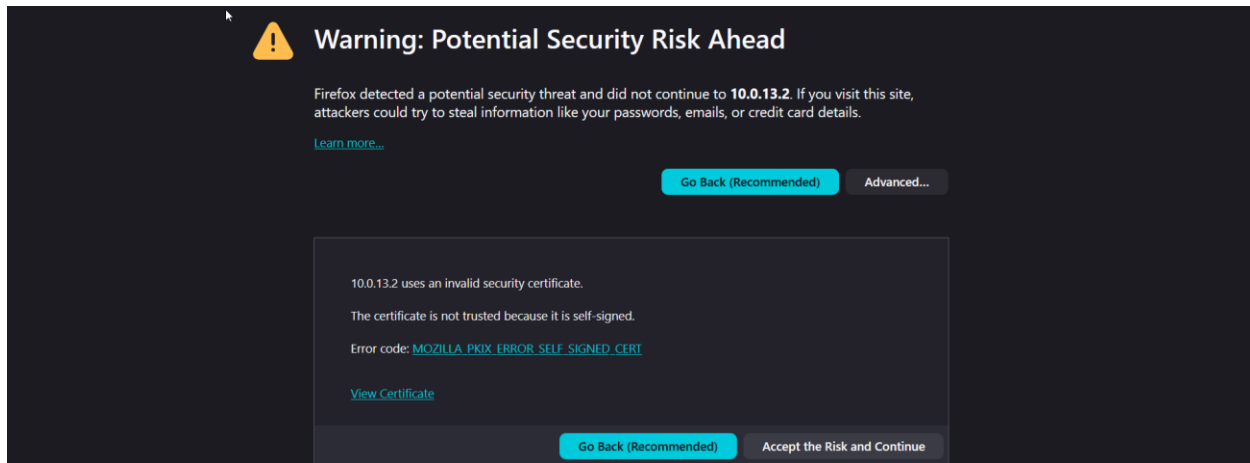


Figure 36: Firefox complaining about Rockstors Self-Signed certificate

It will ask to create a new admin user.

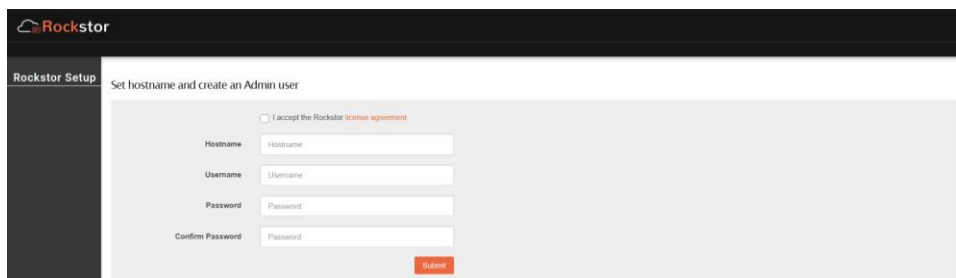


Figure 37: Rockstors initial admin account creation on the webpage

After the user has been created, the Rockstor dashboard can be accessed.

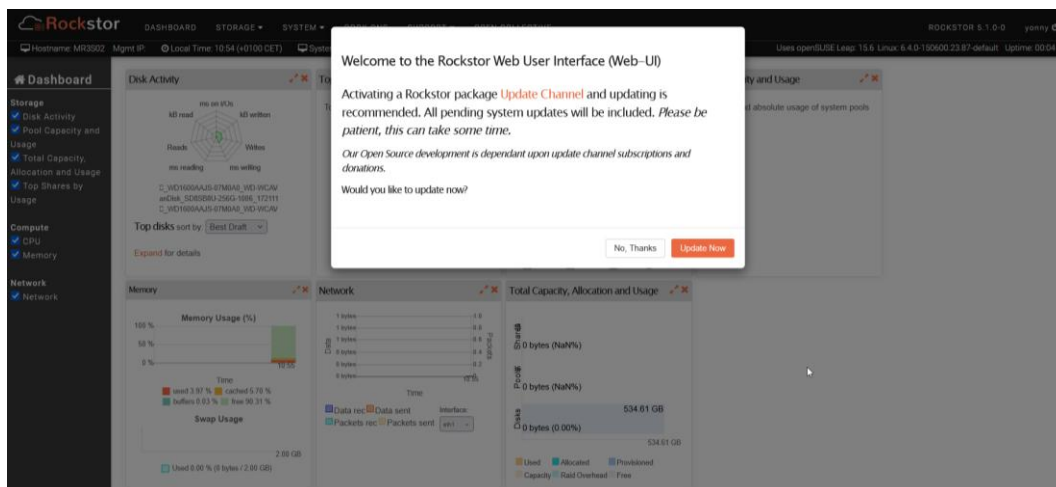


Figure 38: First look of the Rockstors dashboard

RACK INSTALLATION

Initial Rack Installation with Rail installation

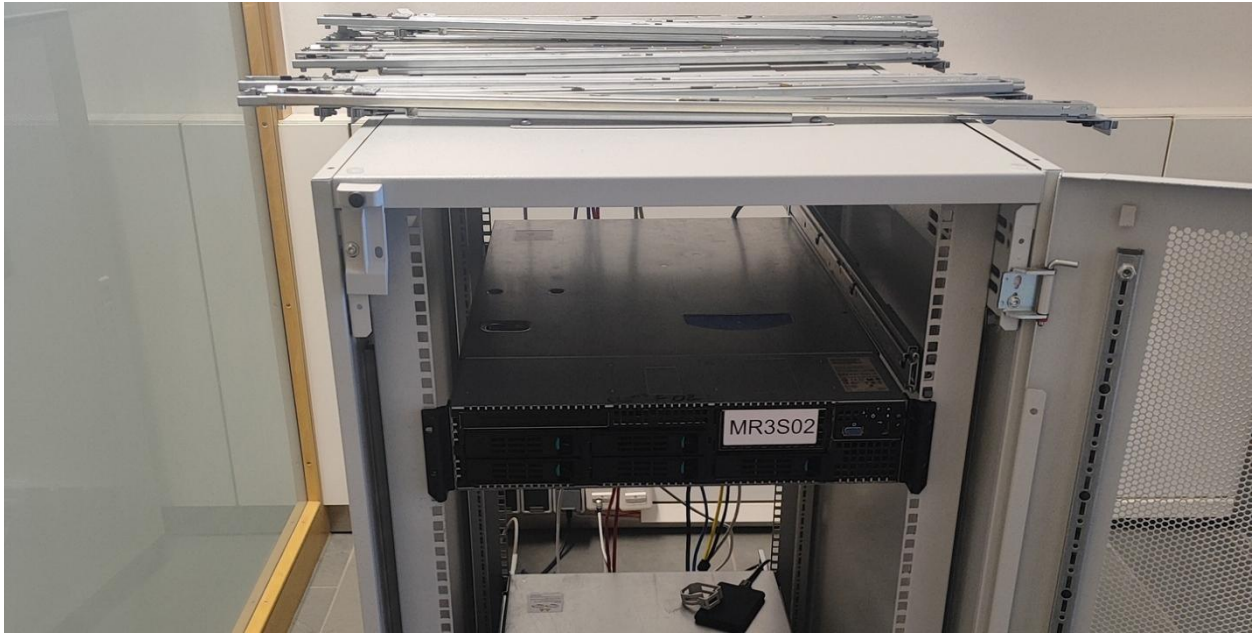


Figure 39: Server installed with rails in the rack

We had decided to stack the Servers by group number, being second we installed one Rail for reference of where the first group would be.

First installation of the Ethernet and power cables

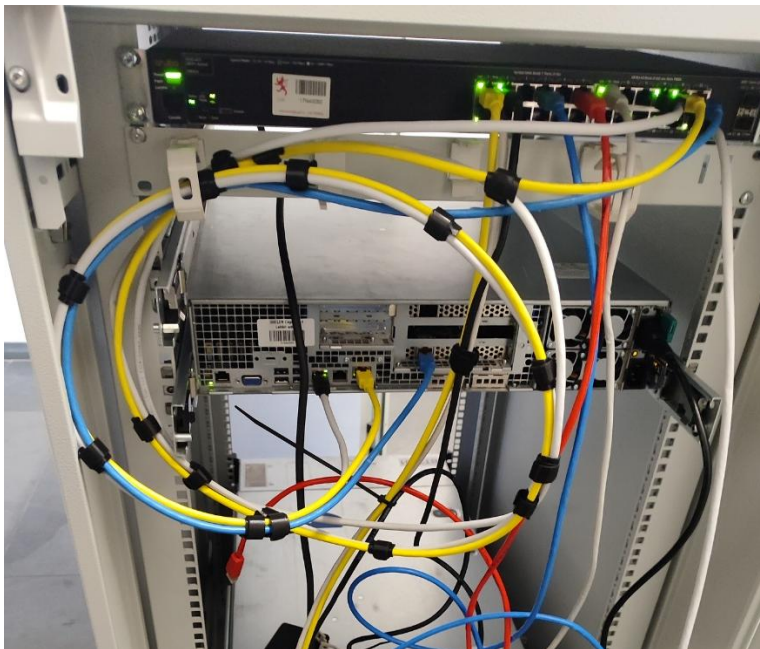


Figure 40: Server connected with power and ethernet

This was not good since the cables were too long to really manage them well so we had just looped them.

Second configuration of the cables, with the help of the 3rd group using shorter cables

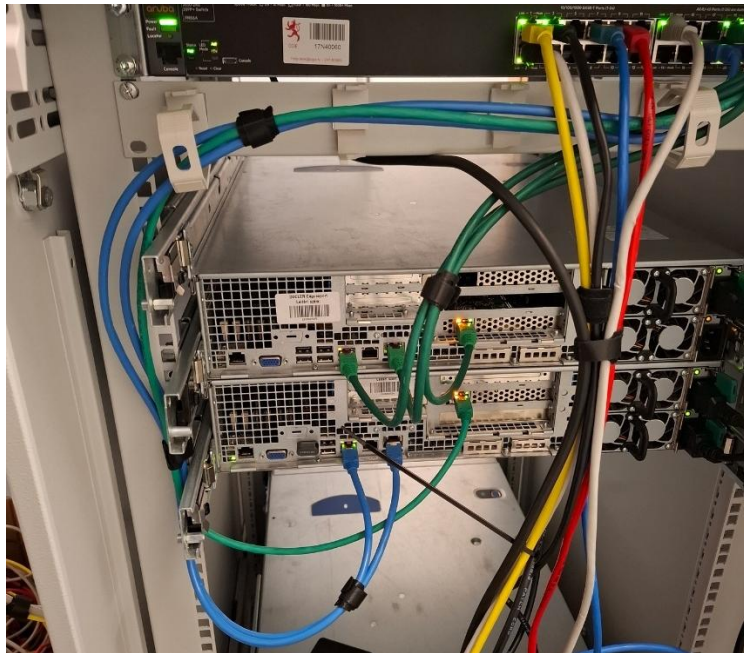


Figure 41: Better cable management with more servers being installed

Final management of the cables with every group

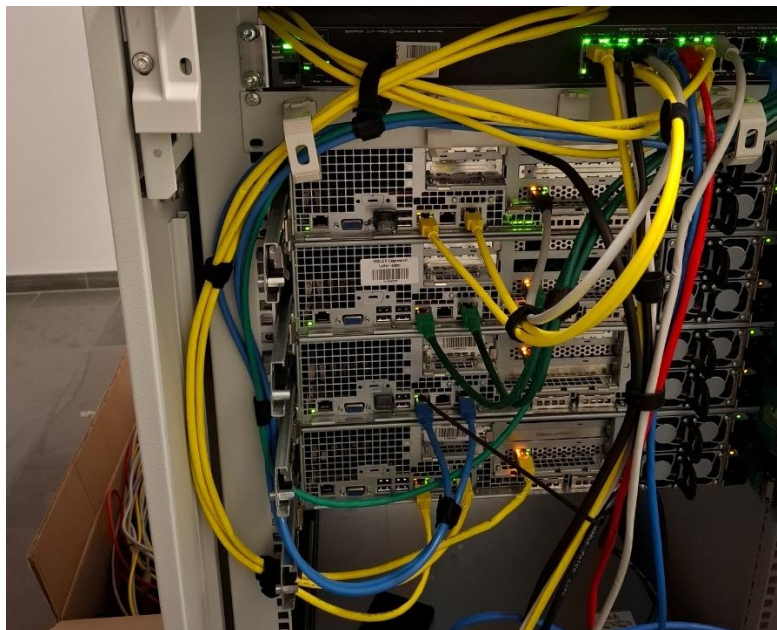


Figure 42: All servers installed

Powering on the Server

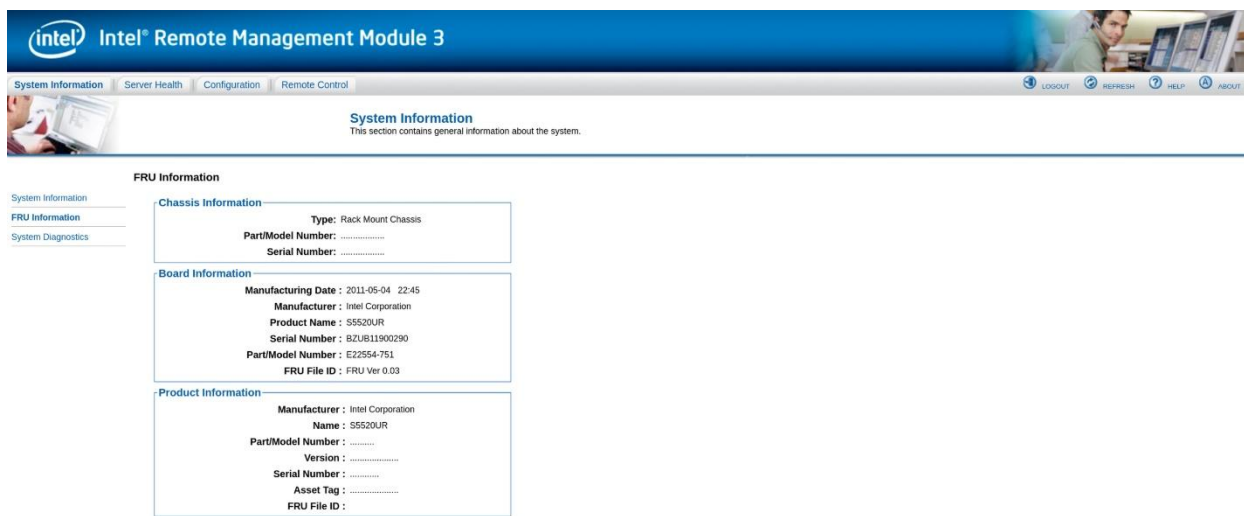


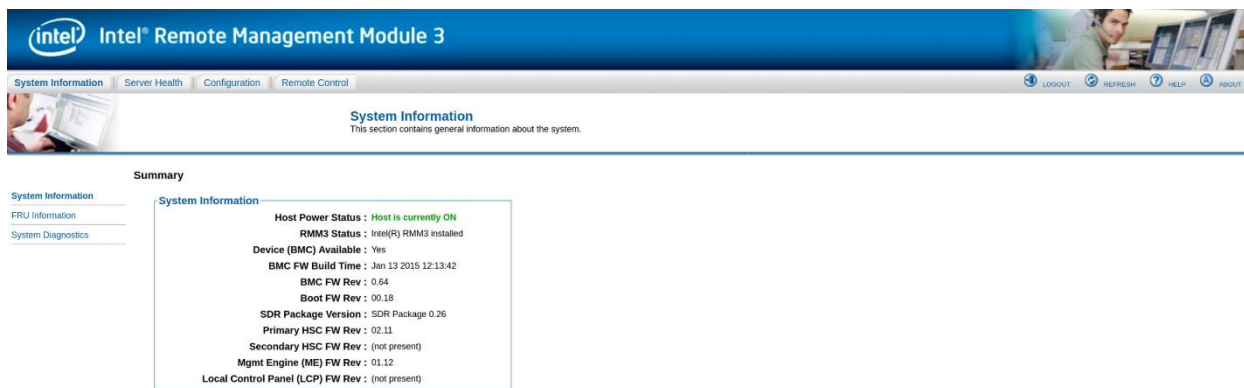
Figure 43: The server powered on

CONNECTION TO RMM



Figure 44: Login page of the servers RMM and the FRU page





System Information
This section contains general information about the system.

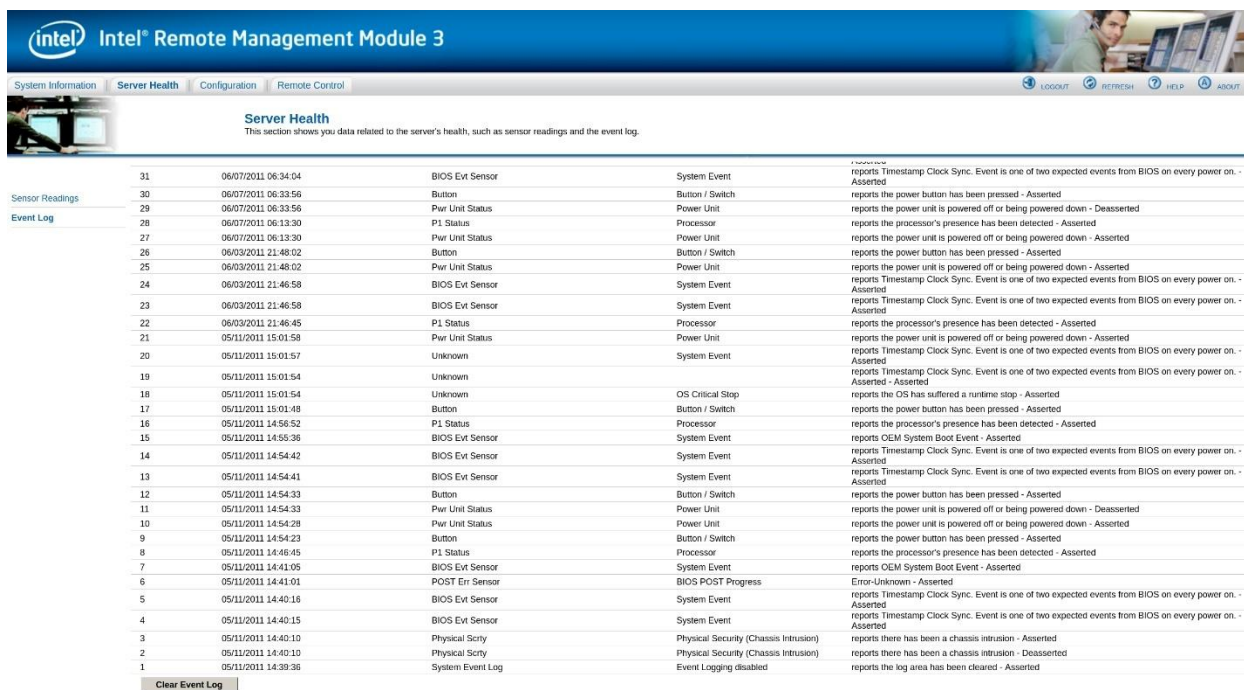
Summary

System Information

- Host Power Status : Host is currently ON
- RMM3 Status : Intel(R) RMM3 installed
- Device (BMC) Available : Yes
- BMC FW Build Time : Jun 13 2015 12:13:42
- BMC FW Rev : 0.64
- Boot FW Rev : 00.18
- SDR Package Version : SDR Package 0.26
- Primary HSC FW Rev : 02.11
- Secondary HSC FW Rev : (not present)
- Mgmt Engine (ME) FW Rev : 01.12
- Local Control Panel (LCP) FW Rev : (not present)

Figure 45: System information on the RMM page

We had an error in the BIOS about a filled error log, in the RMM we found it and cleared it



Server Health
This section shows you data related to the server's health, such as sensor readings and the event log.

ID	Date/Time	Sensor	Event	Description
31	06/07/2011 06:34:04	BIOS Evt Sensor	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
30	06/07/2011 06:33:56	Button	Button / Switch	reports the power button has been pressed - Asserted
29	06/07/2011 06:33:56	Pwr Unit Status	Power Unit	reports the power unit is powered off or being powered down - Deasserted
28	06/07/2011 06:13:30	P1 Status	Processor	reports the processor's presence has been detected - Asserted
27	06/07/2011 06:13:30	Pwr Unit Status	Power Unit	reports the power unit is powered off or being powered down - Asserted
26	06/03/2011 21:48:02	Button	Button / Switch	reports the power button has been pressed - Asserted
25	06/03/2011 21:48:02	Pwr Unit Status	Power Unit	reports the power unit is powered off or being powered down - Asserted
24	06/03/2011 21:46:58	BIOS Evt Sensor	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
23	06/03/2011 21:46:58	BIOS Evt Sensor	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
22	06/03/2011 21:46:45	P1 Status	Processor	reports the processor's presence has been detected - Asserted
21	05/11/2011 15:01:58	Pwr Unit Status	Power Unit	reports the power unit is powered off or being powered down - Asserted
20	05/11/2011 15:01:57	Unknown	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
19	05/11/2011 15:01:54	Unknown	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
18	05/11/2011 15:01:54	Unknown	OS Critical Stop	reports the OS has suffered a runtime stop - Asserted
17	05/11/2011 15:01:48	Button	Button / Switch	reports the power button has been pressed - Asserted
16	05/11/2011 14:56:52	P1 Status	Processor	reports the processor's presence has been detected - Asserted
15	05/11/2011 14:56:38	BIOS Evt Sensor	System Event	reports OEM System Boot Event - Asserted
14	05/11/2011 14:54:42	BIOS Evt Sensor	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
13	05/11/2011 14:54:41	BIOS Evt Sensor	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
12	05/11/2011 14:54:33	Button	Button / Switch	reports the power button has been pressed - Asserted
11	05/11/2011 14:54:33	Pwr Unit Status	Power Unit	reports the power unit is powered off or being powered down - Deasserted
10	05/11/2011 14:54:28	Pwr Unit Status	Power Unit	reports the power unit is powered off or being powered down - Asserted
9	05/11/2011 14:54:23	Button	Button / Switch	reports the power button has been pressed - Asserted
8	05/11/2011 14:46:45	P1 Status	Processor	reports the processor's presence has been detected - Asserted
7	05/11/2011 14:41:05	BIOS Evt Sensor	System Event	reports OEM System Boot Event - Asserted
6	05/11/2011 14:41:01	POST Err Sensor	BIOS POST Progress	Error-Unknown - Asserted
5	05/11/2011 14:40:18	BIOS Evt Sensor	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
4	05/11/2011 14:40:15	BIOS Evt Sensor	System Event	reports Timestamp Clock Sync. Event is one of two expected events from BIOS on every power on. - Asserted
3	05/11/2011 14:40:10	Physical Scry	Physical Security (Chassis Intrusion)	reports there has been a chassis intrusion - Asserted
2	05/11/2011 14:40:10	Physical Scry	Physical Security (Chassis Intrusion)	reports there has been a chassis intrusion - Deasserted
1	05/11/2011 14:39:36	System Event Log	Event Logging disabled	reports the log area has been cleared - Asserted

Clear Event Log

Figure 46: The RMM logs that were full and that we deleted

SHARE CONFIGURATION

We have multiple shares for different purposes on our family NAS. Every family member should have access to their own private space. There is a photo space where only parents should be able to store and look at photos and where other family members can only look at images. There is also a shared space for all family members, with Read and Write permissions. This space is separated on a completely different drive pair.

Our family:

Donny, Yona: Chrëscht

PRIVAT

Only the user who owns the space has access.

Yona(Linux, Windows, SFTP): 10GB

Donny(Windows): 10GB

Chrëscht(SFTP, Android): 5GB

PHOTO

Everyone has ReadWrite access

Entire rest of hardware raid

iSCSI

Possibility of creating network drives on the software raid

Share services we need:

- **SAMBA** (SMB) for Windows
- **Network File System** (NFS) for Linux
- **SSH File Transfer Protocol** (SFTP) for Android
- **iSCSI** for virtual drives

CREATING THE SOFTWARE RAID

Since we don't have any pools yet we create a new pool

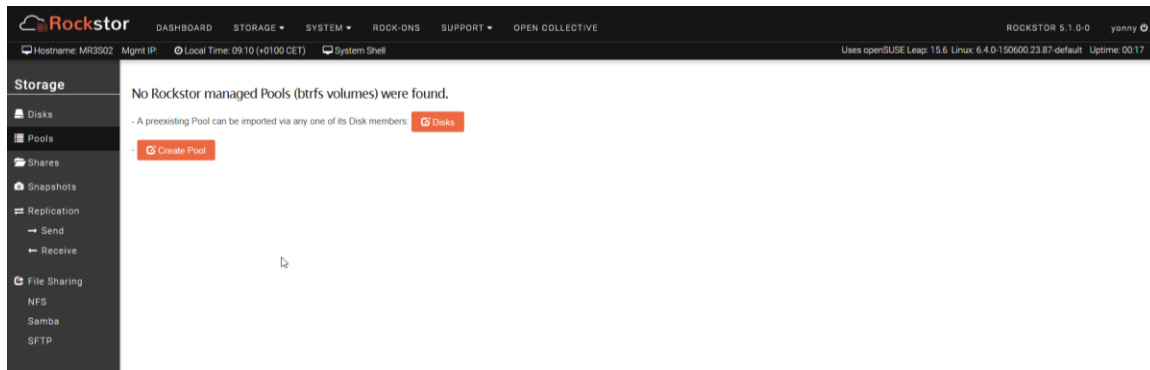


Figure 47: Creating the first pool

We create a new pool with a RAID1 using our 2 HDDs connected to the Mainboard

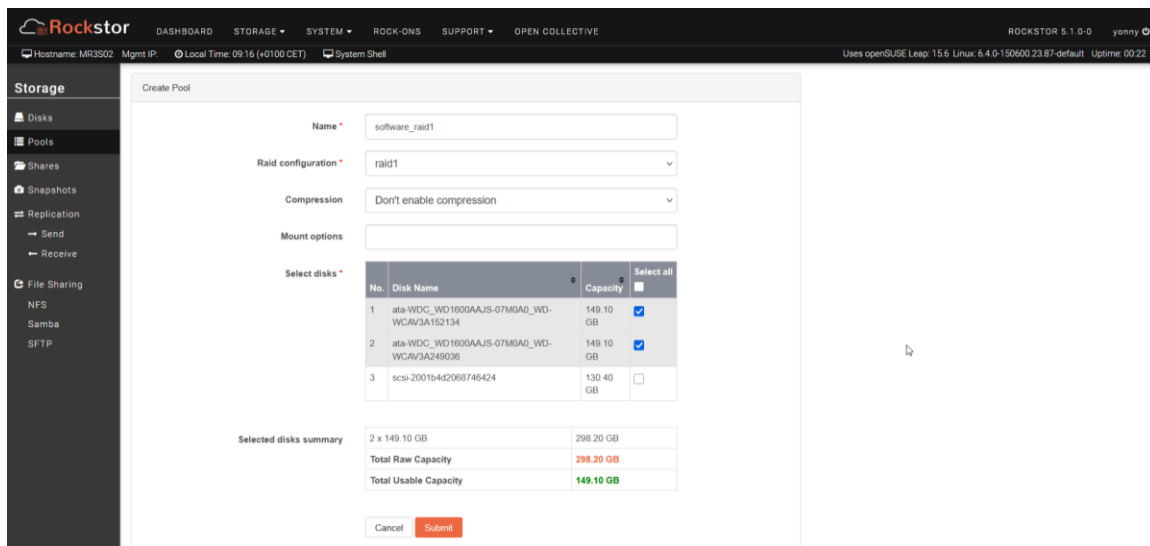


Figure 48: Creating a pool by selecting the 2 physically connected hdds

We verify that the software RAID was successfully created and works

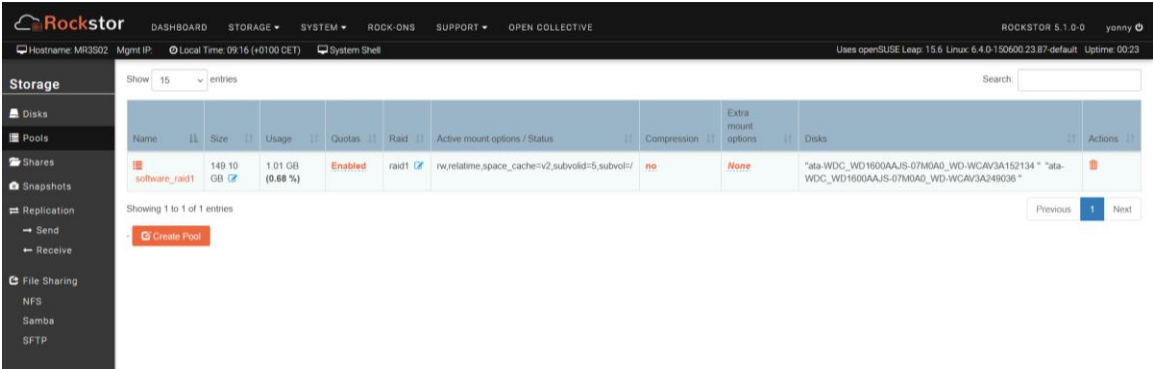


Figure 49: Software storage pool created

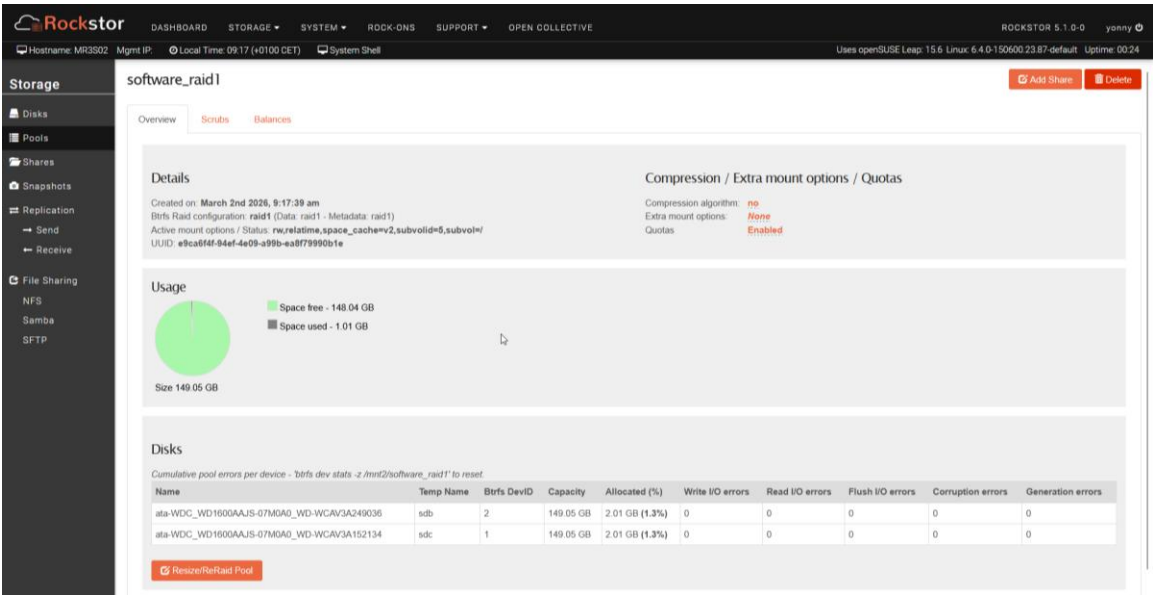


Figure 50: Details of the software raid

CREATING THE USERS AND GROUPS

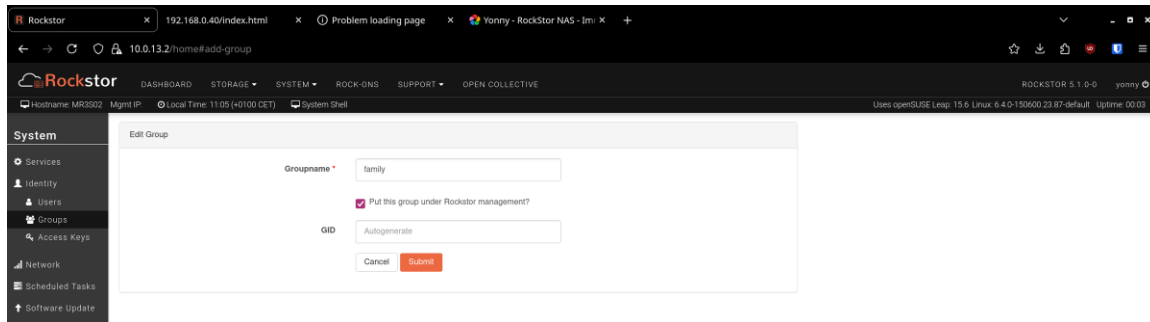


Figure 51: Creating a group for all the users called family

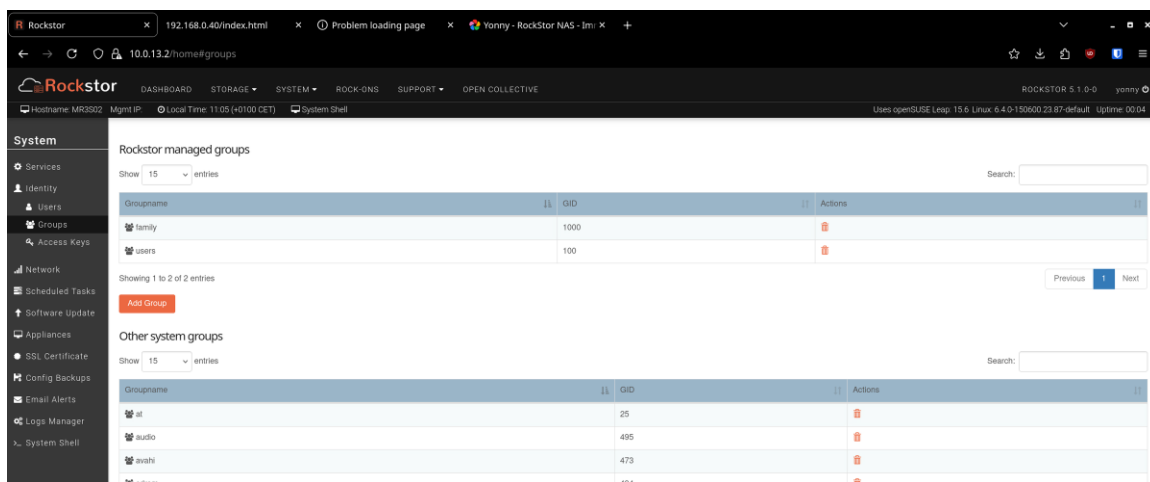


Figure 52: All the groups on the rockstor NAS showing the default "users" group and our newly created "family" group

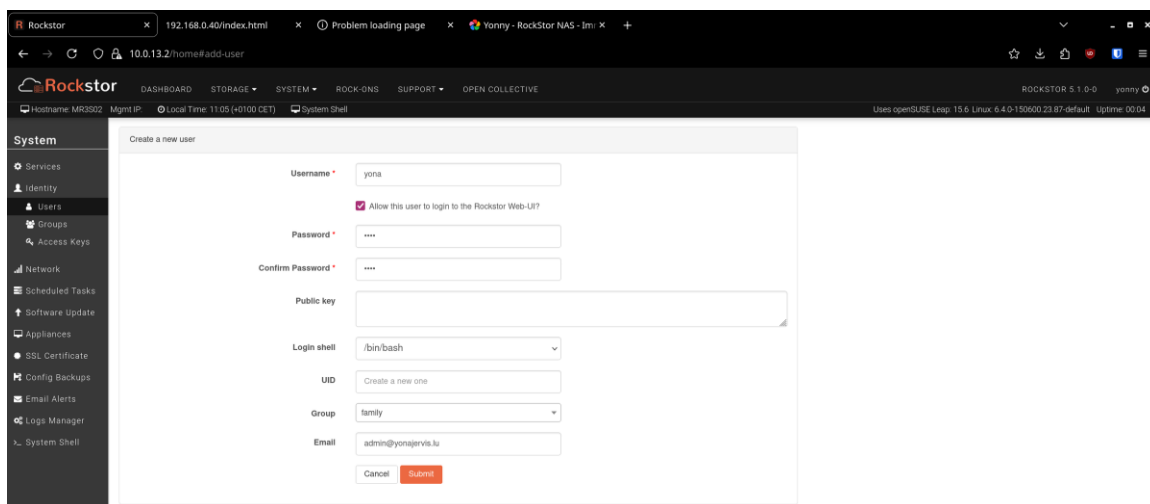


Figure 53: Creating the first user of three

Creating the shares

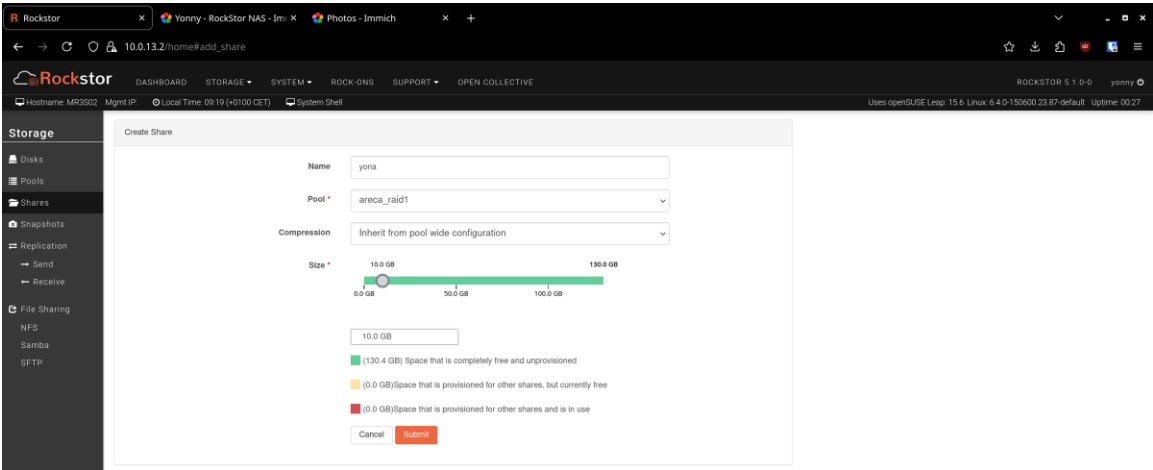


Figure 54: Creating a 10GB share for the "yona" user

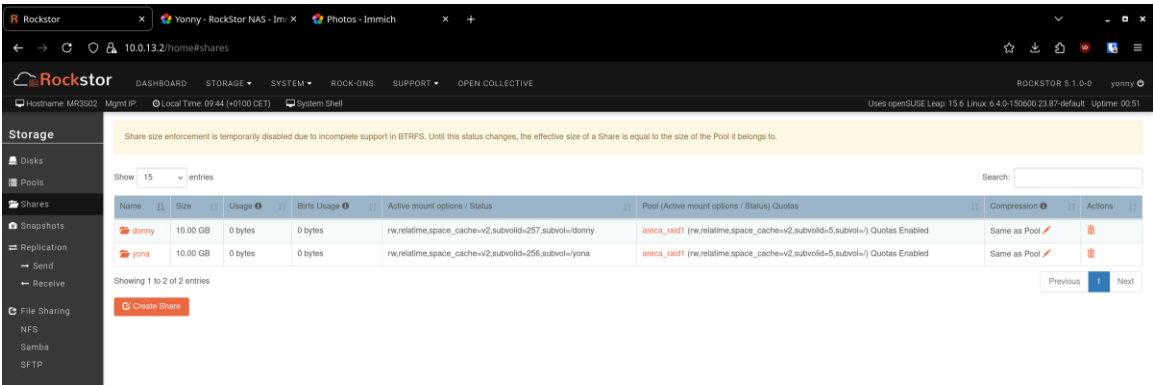


Figure 55: The two private shares for "donny" and "yona" created

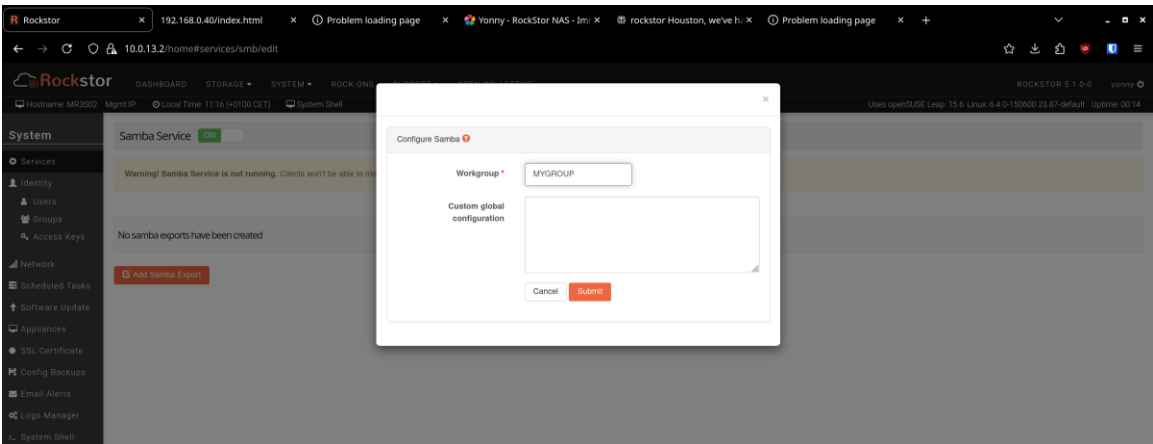


Figure 56: Enabling the samba service with "MYGROUP" as the Workgroup

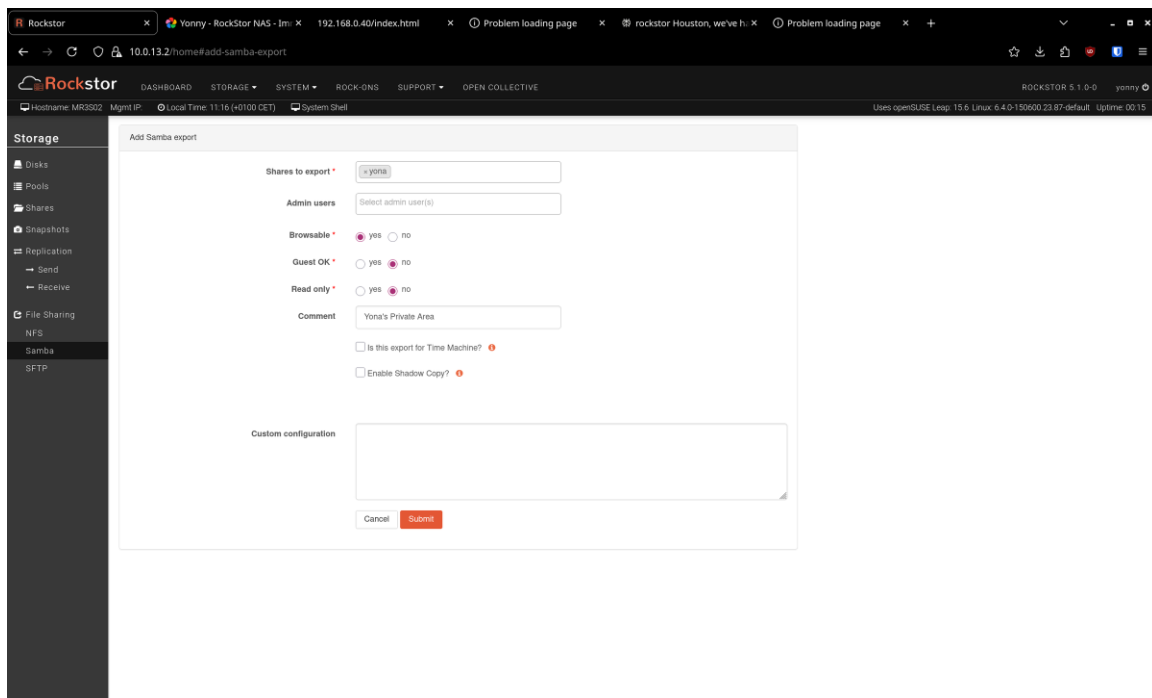


Figure 57: Exporting a private share over SMB

USING THE SHARES

Connecting to the NFS shared folder and using it

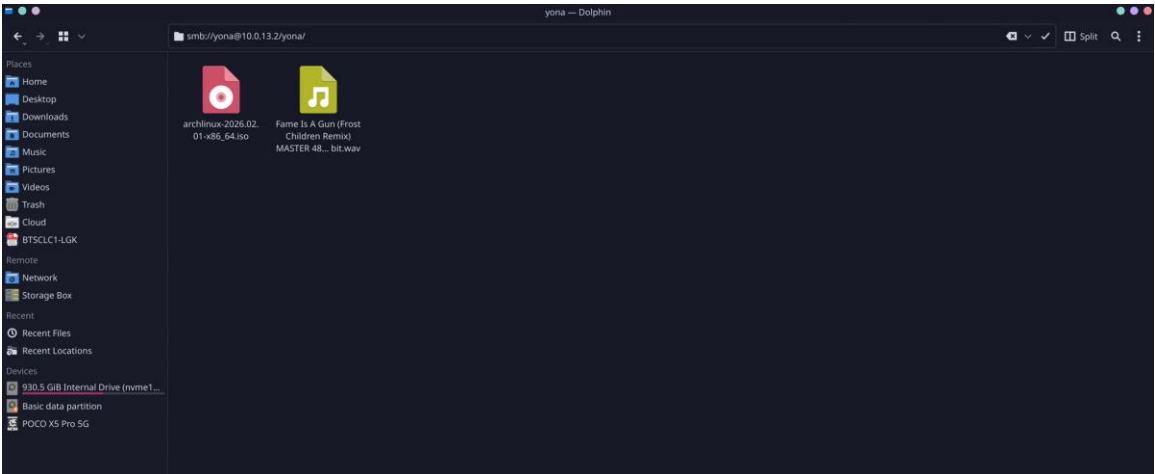


Figure 58: After enabling SMB we can use our credentials to connect to our private share via the IP address

Connecting to the Samba shared folder and using it

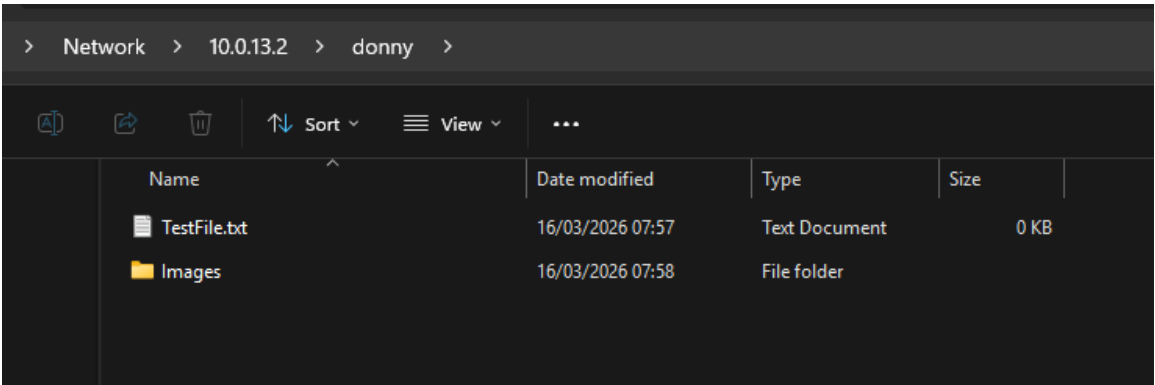


Figure 59: Accessing a private share over Windows

Users using the File shares

Name	Size	Usage	Btrfs Usage
chrescht	5.00 GB	16.00 KB	16.00 KB
donny	10.00 GB	3.96 MB	3.96 MB
photo	100.00 GB	16.00 KB	16.00 KB
shared	148.00 GB	16.00 KB	16.00 KB
yona	10.00 GB	1.49 GB	1.49 GB

Figure 60: All shares listed

ISCSI PROTOCOL

ISCSI, called Internet Small Computer Systems Interface, is a transport layer protocol which uses TCP/IP to access remote storages as if they were a local storage system. So instead of only sharing a folder, it shares the Hard drive, so you can manage the filesystem and partitions.

NFS, called Network File System, is a protocol for sharing files and folders across a network, it's mainly supported for UNIX/Linux systems and where the client requests for a file or folder from the server using remote procedure calls (RPC). The server mounts the file/folder on the client after confirming if it's accessible and the client has the required access permissions.

SMB, called Server Message Block, is also a protocol for sharing files and folders across a network, but now it's mainly used for Windows systems.

Rockstor does not natively support iSCSI, based on a comment on a post in 2022² from the maintainer and project lead, Philip Guyton, iSCSI wouldn't be implemented and would be moved to a BTRFS supported OS. However the OS supports a iSCSI service which would need to be installed.

INSTALLING AND SETTING UP THE ISCSI SOFTWARE

Installing the package targetcli on the server for making an iSCSI instance

```
MR3S02:~ # zypper install targetcli-fb
```

Figure 61: Installing the necessary package for iSCSI

Deleting the “shared” file share and deleting the export on the server to clear the software RAID for sharing it using iSCSI..

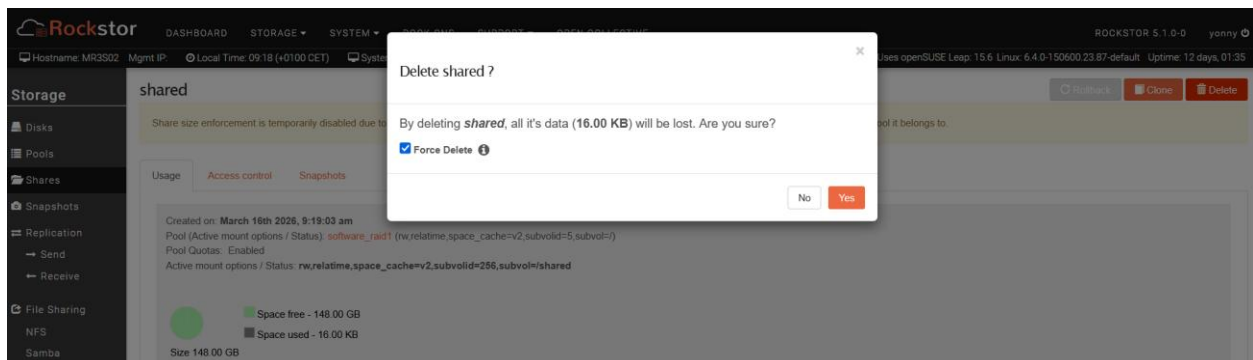


Figure 62: Deleting the temporary share we created to make space for iSCSI

Creating the folder where the image of the iSCSI disk will be stored

```
MR3S02:~ # mkdir /mnt2/software_raid1/iscsi
```

Figure 63: Creating a folder where the iSCSI drive will be mounted

²Source : <https://forum.rockstor.com/t/iscsi-available/8306/2>

Creating the iSCSI disk in the targetcli CLI using the software RAID with 1GB of storage

```
MR3S02:~ # targetcli
targetcli shell version 2.1.54
Copyright 2011-2013 by Datera, Inc and others.
For help on commands, type 'help'.

/> /backstores/fileio create disk01 /mnt2/software_raid1/iscsi/disk01.img 1G
Created fileio disk01 with size 1073741824
/> /iscsi create iqn.2026-03.local.rockstor:disk01
Created target iqn.2026-03.local.rockstor:disk01.
Created TPG 1.
Global pref auto_add_default_portal=true
Created default portal listening on all IPs (0.0.0.0), port 3260.
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/luns create /b
/backstores/fileio/disk01 /bin/ /boot/
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/luns create /backstores/fileio/disk01
Created LUN 0.
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/portals/ ls
o- portals ..... [Portals: 1]
o- 0.0.0.0:3260 ..... [OK]
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/portals/ delete ::0 3260
No such NetworkPortal in configs: /sys/kernel/config/target/iscsi/iqn.2026-03.local.rockstor:disk01/tpgt_1/np/[::0]:3260
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/portals/ delete 0.0.0.0 3260
Deleted network portal 0.0.0.0:3260
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/portals/ create 10.0.13.2 3260
Using default IP port 3260
Created network portal 10.0.13.2:3260.
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/acls create iqn.1996-04.de.suse:01:4cbabf8e3356
Created Node ACL for iqn.1996-04.de.suse:01:4cbabf8e3356
Created mapped LUN 0.
/> saveconfig
Configuration saved to /etc/target/saveconfig.json
/> exit
Global pref auto_save_on_exit=true
Last 10 configs saved in /etc/target/backup/.
Configuration saved to /etc/target/saveconfig.json
```

Figure 64: Opening the iSCSI cli and configuring it the first time

Fixing a mistake where the configuration was made using the IQN of the server instead of the client.

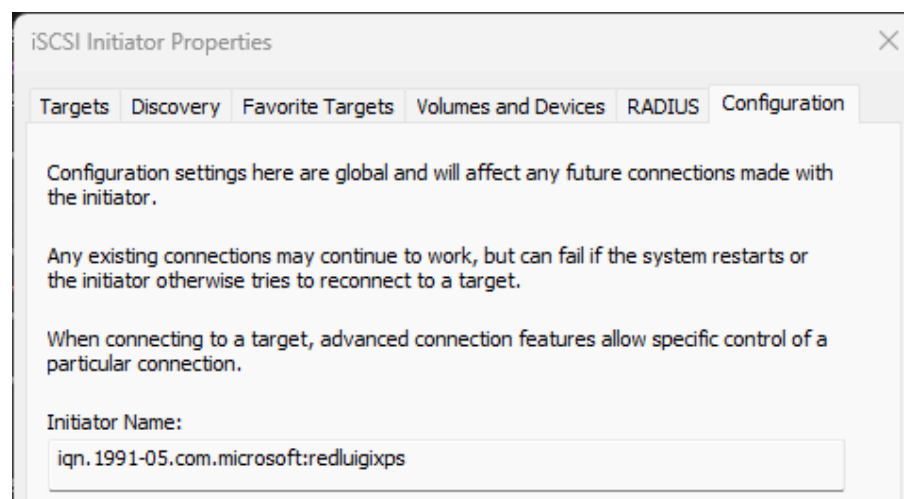


Figure 65: Getting the Initiator ID of the client: Opening the iSCSI initiator properties

```

/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/acls/ delete iqn.1996-04.de.suse:01:4cbabf8e3356
Deleted Node ACL iqn.1996-04.de.suse:01:4cbabf8e3356.
/> /iscsi/iqn.2026-03.local.rockstor:disk01/tpg1/acls/ create iqn.1991-05.com.microsoft:redluigixps
Created Node ACL for iqn.1991-05.com.microsoft:redluigixps
Created mapped LUN 0.
/> saveconfig
Configuration saved to /etc/target/saveconfig.json
/> exit
Global pref auto_save_on_exit=true
Last 10 configs saved in /etc/target/backup/.
Configuration saved to /etc/target/saveconfig.json

```

Figure 66: Configuring iSCSI via the terminal and fixing the remaining issues

After reconfiguring and enabling the service, the client can access the target server iSCSI storage using the servers IP address in the "Quick Connect" section and see the disk in the disk management.

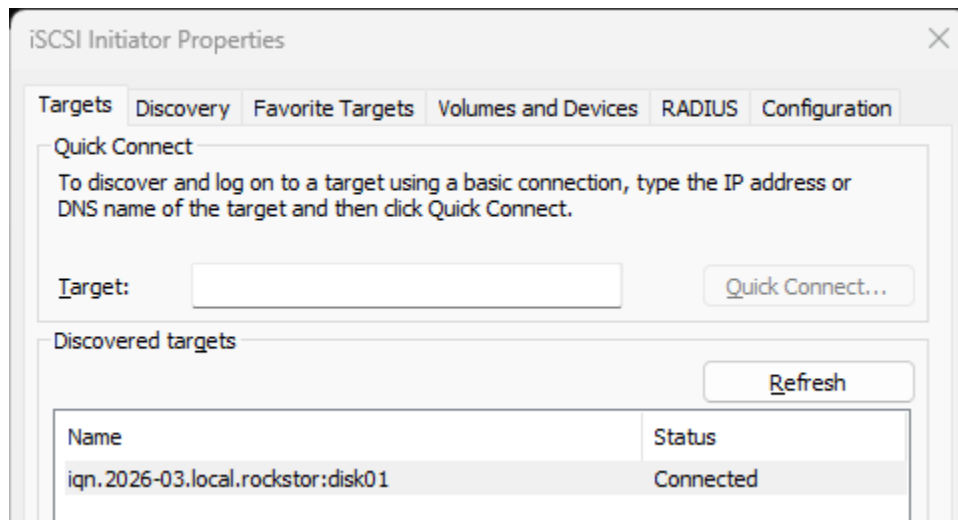


Figure 67: iSCSI Initiator on the client with the newly created VDisk connected

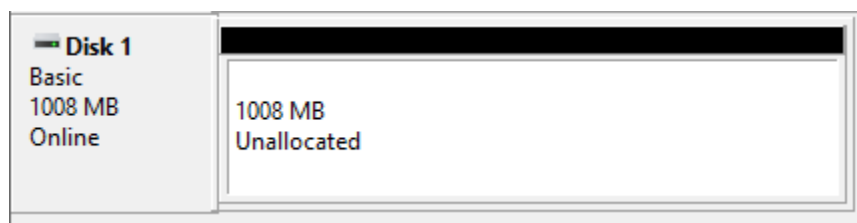


Figure 68: Hard drive showing up like a native drive would in partition manager

Formatting and testing of the iSCSI

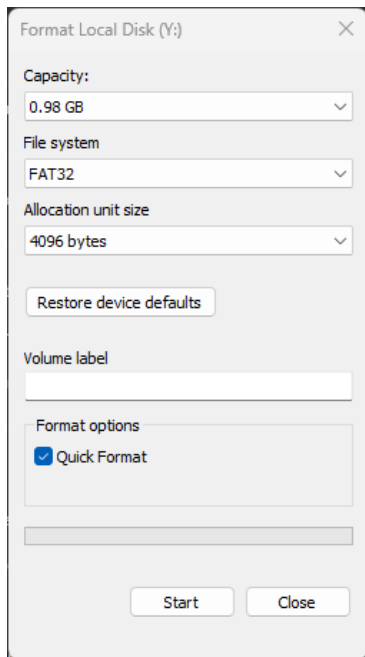


Figure 69: Format the drive

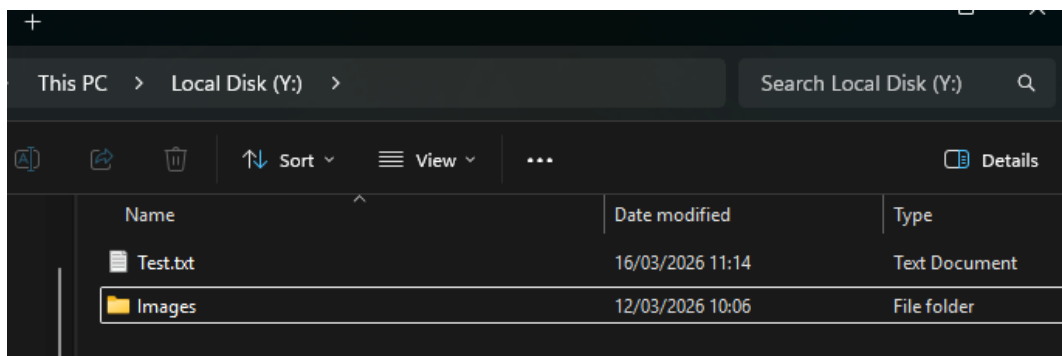


Figure 70: Mount it into your local filesystem